Ruideng Zhong

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Education

Duke University — Durham, NC
BSE in Electrical and Computer Engineering, Minor in Economics — *Expected May 2029*
Cumulative GPA: 4.0/4.0, Major GPA: 4.0/4.0

Experience

Electrical Systems Engineer — Aug 2025 – Present
Duke Motorsports Team (FSAE) — *Durham, NC*

- Designed and fabricated vehicle and dynamometer wiring harnesses using RapidHarness.
- Validated continuity, sensor data, and safety requirements via MoTeC C185 Dash Manager and M1 Tune.
- Used Altium to design schematics and PCBs for the steering wheel, BSPD, strain gauge, and light gates.

Quality Engineering Intern — May 2026 – Aug 2026
EEEA Inc. — *Simpsonville, SC*

- Achieved 98% accuracy in classifying assembled PCBs according to IPC-A-610 criteria.
- Led procedure documentation efforts to support AS9100 certification.
- Reduced AOI false-positive rate by 24% through custom programming and setup.

Molecular Dynamics Simulations Research Intern — Aug 2023 – Aug 2024
Clemson University Department of Electrical and Computer Engineering — *Clemson, SC*

- Adopted molecular dynamics simulations to investigate the material behavior of C60 clusters under compression.
- Optimized C++ programs to reduce runtime by 17% using Clemson's Palmetto 2 HPC cluster.
- Analyzed outputs and logs with Python to validate results and identify anomalies.

Projects

12V to 7V DC-DC Buck Converter PCB | *Altium, Power Electronics, PCBA Assembly* — May 2026 – Present

- Designed a compact 12V to 7V DC-DC buck converter module with an integrated smart high-side switch for regulated power delivery and controlled load switching, with strong thermal performance and power integrity.
- Used Altium Designer to develop the circuit schematic and PCB layout, optimizing signal integrity.

Gate-Level 16-Bit MIPS-Style CPU | *Logisim, MIPS* — Feb 2026 – Apr 2026

- Implemented a 16-bit CPU with 8 registers in Logisim with a custom ALU and a multicycle control unit.
- Designed gate-level registers, counters, and shifters with component constraints.
- Supported arithmetic, logical, memory, and control-flow instructions.

Hardware Acceleration on a Xilinx Zynq-7020 FPGA | *Vivado, Verilog, FPGA* — June 2024 – Aug 2025

- Designed and implemented an FPGA accelerator for real-time image processing using Vivado.
- Developed and pipelined a Sobel edge-detection module to process streaming image data.
- Verified functionality through simulation and hardware testing, achieving a 4x performance improvement.

Technical Skills

Programming and Systems: Java, Python, C, C++, MATLAB, Verilog, MIPS, HTML/CSS, Git, Linux, Unix
Software: Altium, KiCad, OrCad X Capture, OrCad PSPICE, Vivado, RapidHarness, SolidWorks
Laboratory Equipment: Digital Multimeters, Oscilloscopes, Function Generators, Power Supplies, Soldering
Productivity: Microsoft 365 (Excel, PowerPoint, Word), Google Suite, Airtable, Trello

Additional Information

Relevant Coursework: Computer Architecture, Intro to Microelectronics, Signals and Systems, Data Structures and Algorithms, Electricity and Magnetism Physics
Awards: "Most Innovative Design" Award, Pratt Design Expo (2025)
Activities: Duke Marching and Pep Bands (Treasurer), ECE 250 Teaching Assistant (Computer Architecture), Solar Sail Team